

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Branch : Civil Engineering

Semester : V

Subject Code & Name: CV503_Y17_Geotechnical Engineering

Course: T.Y. B.Tech

Max Marks: 60

Date: 11/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	The process of soil compaction results in:	1/CO1	1				
	<table border="1"> <tr> <td>a. Increase in volume</td> <td>b. Decrease in volume</td> <td>c. No change in volume</td> <td>d. Increase in permeability</td> </tr> </table>	a. Increase in volume	b. Decrease in volume	c. No change in volume	d. Increase in permeability		
a. Increase in volume	b. Decrease in volume	c. No change in volume	d. Increase in permeability				
2	In Terzaghi's effective stress principle, the effective stress is equal to:	1/CO2	1				
	<table border="1"> <tr> <td>a. Total stress minus pore water pressure</td> <td>b. Total stress plus pore water pressure</td> <td>c. Pore water pressure only</td> <td>d. Zero</td> </tr> </table>	a. Total stress minus pore water pressure	b. Total stress plus pore water pressure	c. Pore water pressure only	d. Zero		
a. Total stress minus pore water pressure	b. Total stress plus pore water pressure	c. Pore water pressure only	d. Zero				
3	Which of the following is a type of shear test used in soil mechanics?	1/CO3	1				
	<table border="1"> <tr> <td>a. Unconfined compression test</td> <td>b. Triaxial compression test</td> <td>c. Vane shear test</td> <td>d. All of the above</td> </tr> </table>	a. Unconfined compression test	b. Triaxial compression test	c. Vane shear test	d. All of the above		
a. Unconfined compression test	b. Triaxial compression test	c. Vane shear test	d. All of the above				
4	Darcy's law is primarily concerned with:	1/CO2	1				
	<table border="1"> <tr> <td>a. Stress distribution in soil</td> <td>b. Permeability of soil</td> <td>c. Soil compaction</td> <td>d. Soil consistency</td> </tr> </table>	a. Stress distribution in soil	b. Permeability of soil	c. Soil compaction	d. Soil consistency		
a. Stress distribution in soil	b. Permeability of soil	c. Soil compaction	d. Soil consistency				
5	The void ratio is defined as:	1/CO1	1				
	<table border="1"> <tr> <td>a. Volume of water divided</td> <td>b. Volume of solids divided</td> <td>c. Volume of voids divided</td> <td>d. Volume of soil divided by</td> </tr> </table>	a. Volume of water divided	b. Volume of solids divided	c. Volume of voids divided	d. Volume of soil divided by		
a. Volume of water divided	b. Volume of solids divided	c. Volume of voids divided	d. Volume of soil divided by				

	by the volume of solids	by the volume of voids	by the volume of solids	the total volume		
6	The coefficient of consolidation is measured in:				1/CO2	1
	a. m^2/s	b. cm^2/s	c. cm^3/s	d. m/s		
7	Soil classification is done based on:				1/CO1	1
	a. Particle size	b. Atterberg limits	c. Consistency limits	d. All of the above		
8	The degree of saturation in soil indicates:				1/CO1	1
	a. The volume of water present in the soil	b. The amount of air in the soil	c. The proportion of voids filled with water	d. The total volume of the soil		
9	The unconfined compression test is commonly used for:				1/CO3	1
	a. Cohesive soils	b. Cohesion less soils	c. Saturated soils	d. Partially saturated soils		
10	The consolidation of clayey soil is typically controlled by:				1/CO2	1
	a. Vertical stresses	b. Shear strength	c. Pore water pressure	d. Horizontal stresses		
11	In the Mohr-Coulomb failure criterion, the cohesion is represented by:				1/CO3	1
	a. The slope of the failure envelope	b. The intercept on the shear stress axis	c. The stress ratio	d. The angle of internal friction		
12	The standard proctor compaction test is used to determine:				1/CO1	1
	a. The maximum dry density of soil	b. The permeability of soil	c. The shear strength of soil	d. The void ratio of soil		
Q. 2	Solve the following.					12
A)	Define moisture content. Enlist various method of determination of moisture content and explain anyone method in detail.				2/CO1	6
B)	Explain the Following terms:				2/CO1	6

	1) Voids Ratio 2) Porosity 3) Degree of saturation 4) Water Content 5) Specific Gravity 6) Air Content		
Q.3	Solve the following.		12
A)	A soil sample has wet density 21 kN/m ³ and dry density of 19 kN/m ³ . If the specific gravity of soil is 2.60. Calculate the porosity, moisture content and degree of saturation. Assume unit weight of water = 10 kN/m ³	3/CO1	6
B)	What is mean by consistency of soil, Atterberg limit and explain in detail method to calculate the liquid limit of soil.	3CO1	6
Q. 4	Solve Any Two of the following.		12
A)	What are the different factors affecting permeability of soil	2/CO2	6
B)	A 1.20 m layer of the soil (G = 2.65 and porosity = 35%) is subject to an upward seepage head of 1.80 m. What depth of coarse sand would be required above the soil to provide a factor of safety of 2.0 against piping assuming that the coarse sand has the same porosity and specific gravity as the soil and that there is negligible head loss in the sand.	3/CO2	6
C)	Calculate the consolidation settlement for a clay layer given the following: thickness of the layer = 5 m, initial void ratio = 1.2, compression index = 0.3, and the applied stress is 100 kPa.	3/CO2	6
Q.5	Solve Any Two of the following.		12
A)	Explain the fundamental equation of shear strength of soil given by Coulomb. https://www.batuonline.com	2/CO3	6
B)	A cylindrical sample of soil having cohesion of 0.8Kg/cm ² and angle of internal friction of 20°, is subjected to a cell pressure of 1.0Kg/cm ² . Calculate maximum deviator stress at which the sample will fail and the angle made by failure plane with the axis of sample.	3/CO3	6
C)	Derive the equation for effective stress in a soil mass.	3/CO2	6
Q. 6	Solve Any Two of the following.		12

A)	State and explain the assumptions in 'Terzaghi's theory of one-dimensional consolidation'.	2/CO2	6
B)	Explain the difference between Standard Proctor test and Modified Proctor test.	2/CO1	6
C)	A gravity retaining wall retains 10 m of a backfill, $\gamma = 25 \text{ kN/m}^3$ $\phi = 20^\circ$ with a uniform horizontal surface. Determine the total active thrust per meter run of wall. Assume the wall interface to be vertical.	3/CO3	6
*** **			

51643684

51643684

51643684

51643684

51643684

51643684

51643684

51643684

51643684